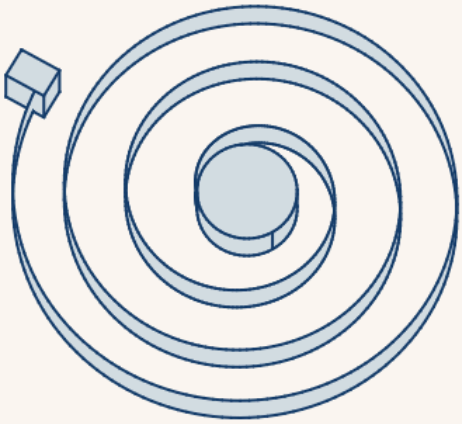
What Every Nerd Should Know About Watches



Every watch you've ever worn has a heart that beats. In a quartz watch it's a sliver of crystal, buzzing 32,768 times a second. In a mechanical watch it's something you can see with your own eyes: a tiny spring, swinging back and forth. Tonight I want to show you why it swings, and why it swings so steadily.

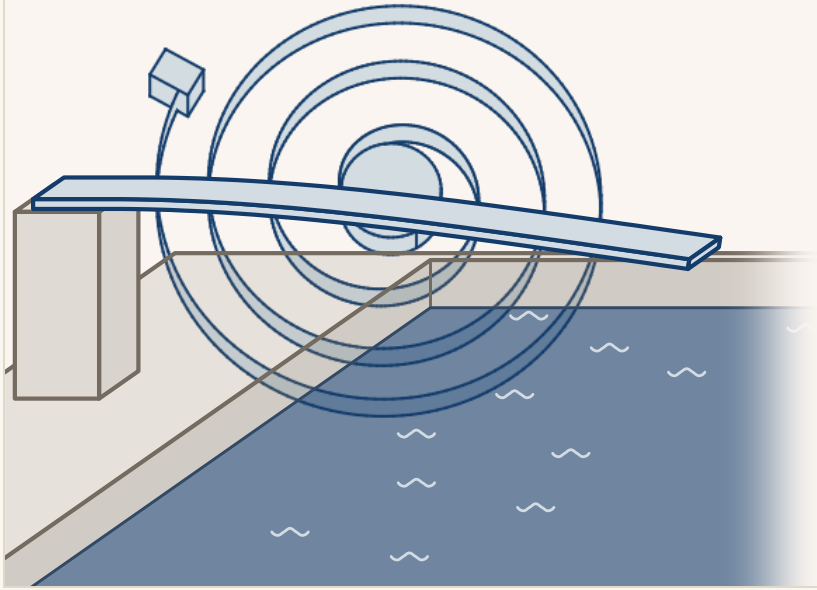
Hairspring and Balance Wheel



This is the heart of a mechanical watch: the *hairspring* and the balance wheel. The spring swings back and forth, one beat every two seconds, and that beat sets the speed of everything else in the watch.

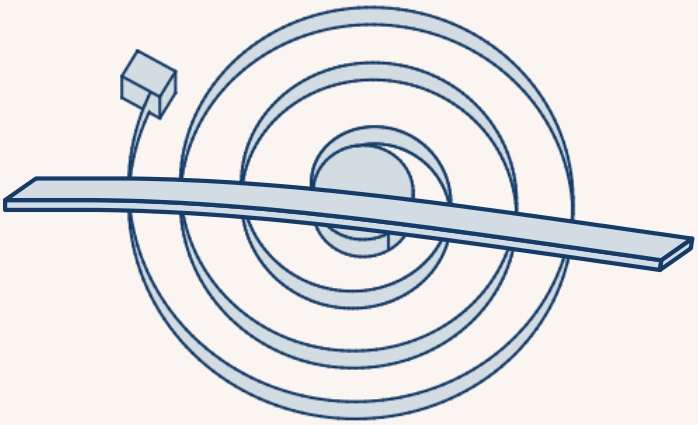
Why does it swing? The spiral makes that hard to see, so let's straighten it out.

Hairspring and Balance Wheel



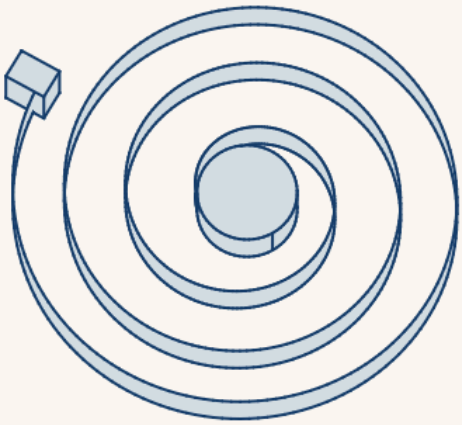
Here is a diving board. Push its end down and it pushes back: that's *elastic restoring force*. Let go, and it doesn't stop at straight. It overshoots, bends the other way, gets pushed back again, and overshoots again. That's an *oscillation*. With no friction, it would bounce like this for ever.

Hairspring and Balance Wheel



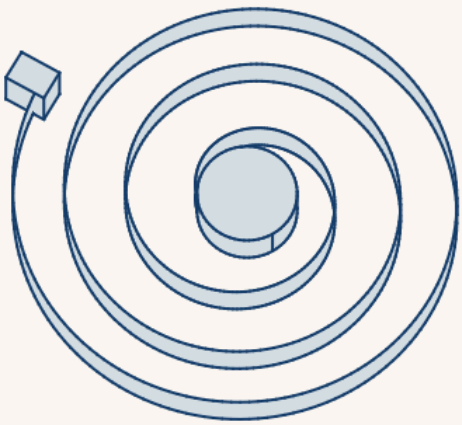
Now let's curl the board up. Its rest shape is no longer a straight line: it's a spiral.

Hairspring and Balance Wheel



And that spiral is exactly the hairspring at rest. Wind it a little tighter and it pushes back towards its rest shape, just as the board pushed back towards straight.

Hairspring and Balance Wheel



So it overshoots, unwinds too far, gets pushed back, and swings, back and forth, just like the diving board.

Before clocks, time was a flow

Before clocks, people told time by watching something flow.

Before clocks, time was a flow



Sundial

±16 min

the Sun vs. a steady clock

c. 1500 BCE

No sun, no time

Sundials go back to about 1500 BCE, and a good one reads to within a few minutes. But no sun, no time. And even a perfect one can be off by up to 16 minutes from a steady clock, because the Sun itself doesn't keep steady time.

Before clocks, time was a flow



c. 1500 BCE

Sundial

±16 min

the Sun vs. a steady clock

No sun, no time



c. 1400 BCE

Water clock

~15 min

per day, the best ones

Freezes in winter

Water clocks work at night, but they slow down as they drain, and they freeze in winter. The best managed about 15 minutes a day.

Before clocks, time was a flow



c. 1500 BCE

Sundial

±16 min

the Sun vs. a steady clock

No sun, no time



c. 1400 BCE

Water clock

~15 min

per day, the best ones

Freezes in winter



9th c. CE

Candle

?

no good data found

Drafts speed it up

Alfred the Great burned four-hour candles, until drafts made them burn out early. How accurate were they? I couldn't find anyone who measured it.

Before clocks, time was a flow



c. 1500 BCE

Sundial

±16 min

the Sun vs. a steady clock

No sun, no time



c. 1400 BCE

Water clock

~15 min

per day, the best ones

Freezes in winter



9th c. CE

Candle

?

no good data found

Drafts speed it up



by 1338

Hourglass

?

no good data found

Someone must flip it

Hourglass sand flows at a constant rate and doesn't mind a rolling ship. But someone has to flip a half-hour glass 48 times a day.

Before clocks, time was a flow



c. 1500 BCE

Sundial

±16 min

the Sun vs. a steady clock

No sun, no time



c. 1400 BCE

Water clock

~15 min

per day, the best ones

Freezes in winter



9th c. CE

Candle

?

no good data found

Drafts speed it up



by 1338

Hourglass

?

no good data found

Someone must flip it

Their rate is set by the outside world. A clock sets its own.

Every one of these runs at a rate set by the outside world. A clock sets its own. Remember 15 minutes a day: the first mechanical clocks probably didn't beat it.